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Lessons from a next generation carbon ranching experiment

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ABSTRACT

Identifying grazing practices that maximize food production and soil carbon sequestration is of global importance. Currently, we cannot reliably predict which grazing practice will accrue the most soil organic carbon (SOC) partly because of confusion created by an abundance of low-quality experiments. Data quality has likely suffered because adequate treatment replication requires a lot of land, meaningful scale introduces heterogeneity among replicates, and pre-existing differences are often not taken into account. Here we tested the effects of five grazing treatments on SOC stocks with a gold standard randomized controlled trial (RCT) with pre-treatment data. Treatments were no grazing, severe summer grazing, moderate fall grazing, severe fall grazing, and the conventional approach for the system of moderate summer grazing. Because grazing experiments are often little-replicated, we also gauged the dependence of results on pre-treatment data by comparing the results of analyses with vs without pre-treatment data. We also tested for relationships between plant (root biomass, plant species composition) and soil properties (mass, nitrogen concentration) and SOC. After applying treatments for 5-yr, the no grazing and severe summer and fall grazing treatments accrued 0.85 to 1.22 kg $\times$ m^{-2} (0–60 cm soil depth) more SOC than conventional moderate summer grazing and represent appreciable (i.e. >4 per mille per yr) increases in SOC stocks. Furthermore, the entire portfolio of new grazing practices increased SOC accrual compared to conventional management. Given the divergent characteristics of the newly applied practices, sustainable accrual of SOC may depend on management heterogeneity. Accrual of SOC was likely driven by a reduction in SOC mineralization related to increased soil density and nitrogen, decreased root biomass, and change in plant species composition. Our findings counter the prevailing view that plant material inputs are the primary driver of SOC accrual in grazing lands. In other words, SOC accrual was mainly associated with indicators of decreased SOC mineralization (C outputs). We also found unique results for analyses with and without pre-treatment data, thereby indicating the risk of unreliable results for little-replicated RCTs without pre-treatment data. Robust generalizations on grazing practices that will reliably and sustainably maximize the rate of SOC accrual for a specific system, climate and timeline are still lacking, but further focus on factors affecting SOC mineralization will facilitate their development.

1. Introduction

To reduce climate change and feed growing populations, scientists are looking for ways to optimize agricultural practices for both increased food production and carbon (C) storage (Harrison et al., 2021; Lal et al., 2004; Paustian et al., 2016; Searchinger et al., 2018). Since grazing lands are the most widespread terrestrial biome in the world (Ellis and Ramankutty, 2008), even small increases in soil organic carbon (SOC) stocks (mass SOC $\times$ area^{-1}) will result in appreciable C sequestration. Companies (e.g. 3Degrees, BCarbon, Grassroots Carbon, NativeEnergy, Regen Network Development, Soil Value Exchange, Verra) have started C offset programs where ranchers are paid to modify their grazing management to increase SOC stocks.

Unfortunately, according to recent syntheses (National Academies of Sciences and Medicine, 2019; Reinhart et al., 2022b), it is unclear whether this practice of “C ranching” actually increases SOC stocks. Evidence for the effectiveness of C ranching consists of poorly replicated studies of overly simplistic grazing treatments (grazed vs not grazed)

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that lack pre-treatment data and rely on problematic SOC stock metrics (e.g. Reinhart et al., 2022b). A next generation of grazing experiments are urgently needed to overcome these limitations of the existing research (Chang et al., 2018; FAO, 2019; Gosnell et al., 2020; Reinhart et al., 2022b; Reinhart et al., 2021).

In the current study, we performed a well-replicated study with realistic grazing treatments that avoided problematic SOC stock metrics in the Northern Great Plains, USA. We tested the effect of conventional management vs new management on change in SOC stocks over a 5-yr period. Our experiment included five grazing management treatments (i.e. summer grazing with moderate forage utilization [i.e. conventional], summer with severe utilization, fall with moderate use, fall with severe use, and no grazing). Only one other study (Schönbach et al., 2012) has tested the effect of grazing treatments on SOC stocks with a randomized controlled trial with pre-treatment data, and no other study has tested the effects of grazing season. We predicted a decrease in SOC stocks for paddocks grazed severely (or not grazed) compared to moderately grazed paddocks (Reinhart et al., 2021; Schuman et al., 1999) but an increase in SOC stocks by grazing during the fall. Secondarily, we measured root biomass, soil bulk density, soil nitrogen concentration, and plant species composition. We then tested for relationships between change in those properties and change in SOC stocks (Fig. 1). Stocks may increase by either increasing soil nitrogen (e.g. Piñeiro et al., 2009; Roy and Bagchi, 2022), soil density (Reinhart et al., 2021), root biomass (Wilson et al., 2018), or shifts in plant species composition (e.g. Bagchi and Ritchie, 2010) (Fig. 1). To gauge the dependence of results on pre-treatment data, we used our novel dataset to compare the results of analyses with and without pre-treatment data. Although randomized controlled trials (RCT) are often described as the gold standard, we hypothesized that RCT may produce unreliable results when little-replicated and compared to RCT with pre-treatment data.

2. Materials and methods

2.1. Study site and grazing experiment design

The experiment was centrally located in the Northern Great Plains (105°57′59.1″W, 46°24′24.8″N) and in a pasture historically grazed at moderate intensity in summer. This region supports an estimated 11 million animal unit months of livestock grazing (Reinhart and Vermeire, 2017). The experiment was conducted at a calcareous grassland site with a Pinehill loam soil (United States Department of Agriculture, 2003). The grassland is one of the most common regional grassland types (Martin et al., 1998) and was dominated by the C_3 graminoids *Hesperostipa comata* (31 %), *Pascopyrum smithii* (17 %), *Carex filifolia* (12 %) and the C_4 *Bouteloua gracilis* (12 %).

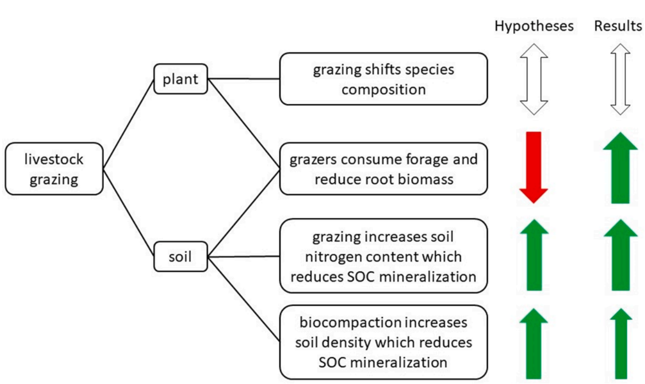

Fig. 1. Hypotheses and results of the vegetation and soil properties affected by livestock grazing treatments that may affect the accrual of soil organic carbon (SOC). Downward red arrow indicates less SOC accrual, whereas upward green arrows indicate greater SOC accrual. Bi-directional arrows indicate multivariate relationships where different plant species may have divergent associations with SOC accrual. The arrow size indicates the relative importance of the process in influencing the accrual of SOC. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The 5-yr experiment included five grazing treatments (i.e. no grazing, severe summer grazing, moderate fall grazing, severe fall grazing, and the conventional moderate summer grazing). Grazing intensity treatments (moderate, severe) were factorially combined with two grazing seasons (June/July, October/November) and applied to 20 total paddocks (60 × 30 m) arranged in a grid in a randomized complete block design with 5 replications (Fig. 2B of Reinhart et al., 2022b). Fort Keogh Livestock and Range Research Lab's Institutional Animal Care and Use Committee evaluated our experiment and determined that our use of animals was consistent with standard livestock management and did not require special approval. In the first year (2013) of the experiment, summer grazing was in August. Cattle (*Bos taurus*) grazing intensities approximated recommended (i.e. moderate; 1 animal unit month [AUM] × ha^{-1}) and heavy (1.5 AUM × ha^{-1}) stocking rates, with an estimated 656 and 309 kg × ha^{-1} of post-grazing residual standing biomass, respectively. No grazing plots (fenced 4.8 × 4.8 m) were established (2013) before the pasture was annually grazed and 20 to 48 m from the grid of paddocks.

2.2. Field sampling

We randomly sampled soil from five of 25 points of a microsite (4 × 4-m grid with points every 1 m) in each paddock and no grazing plot prior to treatment between June 26 and July 8, 2013 (Ellert et al., 2002). Then we repeated this sampling at five different points in the grid from May 31 to June 21, 2018. In each sampled year, we collected 375 samples (5 grazing treatments × 5 treatment replications × 5 cores per experimental unit × 3 depth increments per core = 375). Soil was cored with a column cylinder (87 mm internal diameter) with a removeable side and hand-held jackhammer (Reinhart and Vermeire, 2017). Compaction of the soil column during the sampling procedure was corrected for based on the measured length of the soil core and the borehole depth (Don et al., 2007). Compaction averaged 1.5 % ± 2.6 in 2013 and 1.1 % ± 1.4 in 2018 (mean ± standard deviation). The soil cores were segregated into 0–10, 10–30, and 30–60 cm increments and placed in sealed plastic bags that were transported in coolers back to the lab. Samples were frozen until processed further.

Plant species composition was determined with line point-intercept transects immediately before the June grazing period in 2014 and 2018. Basal and aerial cover for each species were measured using four permanent 5-m transects that were randomly placed in each grazed paddock and read at 20-cm intervals. Current-year or older dead material were distinguished to estimate percent current-year biomass. We were unable to measure transects pre-treatment in 2013. However, plant species composition is known to shift gradually in response to a change in grazing management (Porensky et al., 2016), and data collected in 2014 function as pre-treatment data.

2.3. Soil analysis

Our primary aim was to measure SOC stocks, and we were secondarily interested in measuring predictor variables (e.g. root biomass, soil nitrogen [N]) of variation in stocks. Soil samples were thawed, broken up, air-dried to constant weight, and weighed. Then a 5-g subsample from each sample was further dried (105 °C, 24 hr) then weighed to determine % dry matter.

Roots were extracted from air-dried samples as described by Reinhart and Vermeire (2016). In brief, samples were sieved with a 4.75-mm then 2.0-mm sieve and roots removed by hand. Then, to quantify very fine roots, remaining soil roots were sorted by hand from a 10-g subsample. Roots were dried to constant mass (105 °C for 1 day), weighed, combusted at 550 °C for 16 hr, and reweighed to determine the % ash.

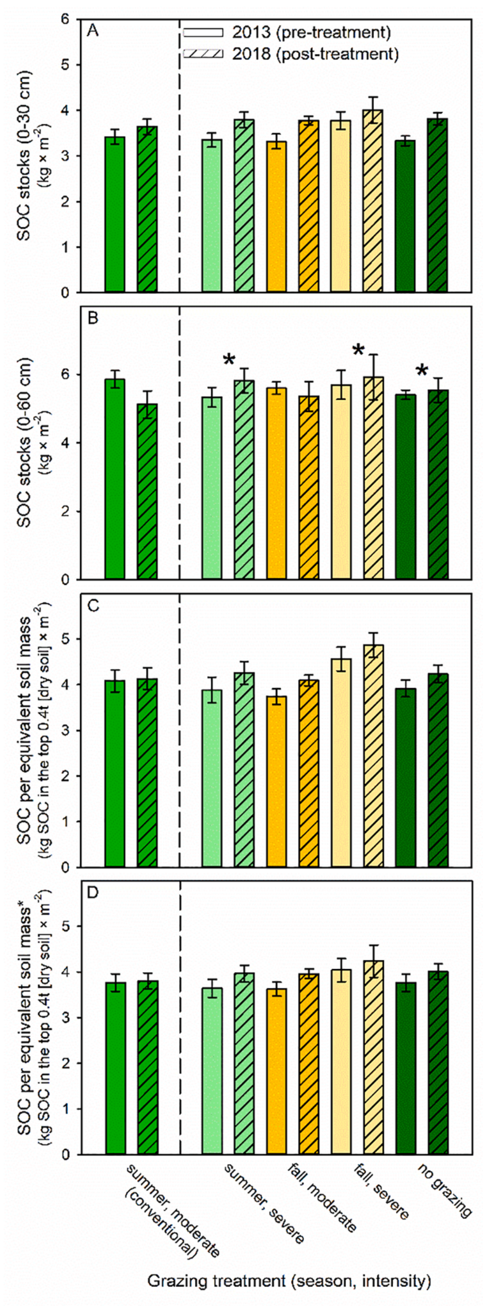

Fig. 2. Effect of grazing treatment (season, intensity) and year on soil organic carbon (SOC) stocks (±1 SE) of a field experiment with five replications in a Northern mixed-grass prairie. Prior to 2013, all experimental units were conventionally managed by livestock (i.e. moderately grazed in summer). SOC stock estimates were for fixed depth increments (A, B) and an equivalent mass of dried surface soil which was corrected for roots (C) (standard metric by Gifford & Roderick (2003)) or roots and rocks (D). Bar cross hatching indicates the sampling year (2013 not hatched; 2018 hatched) and distinguishes whether data were pre- or post-treatment. Significant differences ($P \leq 0.05$) in change in SOC stocks between conventional management vs other treatments are indicated with asterisks. Full statistical results are reported in Table 1.

For each core sample, we summed the total root dry masses for the three material size classes. Rocks (>2 mm) were also collected by hand and weighed.

To determine SOC stocks, we measured each sample's total carbon and soil inorganic carbon (SIC) concentrations using methods of Reinhart and Vermeire (2017). In brief, from the 10-g root-free subsample, we removed and ground 1.5 g of soil in square bottles on a roller-milling device. The SOC concentration (%) of ground samples was determined by measuring total carbon concentration with the dry combustion method (Flash 2000 Series Carbon-Nitrogen Thermo Combustion Elemental Analysis system, Thermo Scientific, MA, USA) and then subtracting the SIC, which represents the carbonate-C. The SIC was determined by the modified pressure-calcimeter method in 2013 samples (Sherrod et al., 2002), and a modification of this method was used to assay the 2018 samples (Stetson and Osborne, 2015). Total nitrogen concentration was measured with the dry combustion method. Total carbon, nitrogen, and SIC were adjusted for % dry matter (105 °C, 24 hr). The SOC values were used to estimate multiple measures of SOC stocks (mass SOC × area^{-1}). Four stock measures were based on two soil depth increments (0–30 or 0–60 cm) with two types of soil (i.e. rock-free fine soil, rock- and root-free fine soil). Unfortunately, grazing treatments affect bulk density (Byrnes et al., 2018), and estimated effects of grazing on SOC stocks for a fixed soil depth are likely to be biased when treatments affect soil bulk density (Rovira et al., 2015). One approach to limiting the impact of compacted surface soils is simply to assay a larger than recommended (>30 cm depth) increment of soil (Reinhart et al., 2022b). Furthermore, we estimated SOC stocks for an equivalent mass of soil (i.e. kg SOC in the top 0.4 t (dry soil) m^{-2}) following Gifford and Roderick's (2003) equation 4 for root-free soil and another for root- and rock-free soil. We provide the full dataset and R script for data manipulation and analysis (Reinhart et al., 2022a).

2.4. Data analysis

We tested the effect of grazing treatments on change in SOC stock metrics. We used R (R Development Core Team, 2020) and *lme4* (Bates et al., 2015) to perform a linear mixed effects analysis of the relationship between grazing treatment and year on SOC stock metrics. As fixed effects, we entered grazing treatment and year (with interaction term) into the model. As a random effect, we included paddocks (and no grazing plots). We used contrasts to test effects of conventional management vs either 1) all other management types or 2) each other management type separately. For each dataset of 250 values (5 grazing treatments × 5 experimental units [paddocks] × 2 years × quintuplicate samples per paddock per year), we used the Rosner's test for outliers in *EnvStats*. We removed between one to seven outliers per dataset. After removal of outliers, visual inspection of residual plots did not reveal any obvious deviations from homoscedasticity or normality. P-values were obtained by likelihood ratio tests of the full model including the interaction term against the model without the interaction term.

To gauge the dependence of results on pre-treatment data, we compared the results of analyses with (described above) vs without pre-treatment data. With only the post-treatment data, we performed routine tests of the effect of conventional management treatment vs each other grazing treatment with a linear model using contrast comparisons. Specifically, we used the full dataset with outliers removed (see above), subset the data collected after applying treatments (2018), averaged data per experimental unit (paddock or plot), and conducted routine analyses.

We also tested for relationships between change in SOC stocks and change in plant (root biomass, species composition) and soil (bulk density, nitrogen) properties of grazed areas (i.e. paddocks) (Fig. 1). First, we averaged the soil data per paddock and year. Correlation coefficients among all plant and soil properties are also provided (Table 2). To identify the best minimum dataset for predicting change in SOC stocks per m^2 (0–60 cm and equivalent soil mass [corrected for roots and rocks]), we performed multiple linear regression (MLR) analyses using plant and soil properties. First, we estimated change in soil properties (soil mass, root biomass, and N concentration) using log response ratios ($\ln[\text{variable}_{2018}/\text{variable}_{2013}]$). In addition, we performed a principal component analysis (PCA) of plant community composition data in *vegan* (Oksanen et al., 2015) in R. Plant abundance data were transformed with the command decostand(hellinger) in *vegan*, and total inertia was 0.25. To estimate change in community composition between 2014 and 2018, we determined the absolute difference in PCA site scores per paddock. We derived two estimates of compositional change. The first was based on PC1 (eigenvalue = 0.11, proportion explained = 0.44), and another summed the change for PC1 and PC2 (eigenvalue = 0.05, proportion explained = 0.22). MLR predictor variables of change included: plant community composition data (two metrics), soil mass (0–60 cm depth), nitrogen (0–30, 0–60 cm), and root mass (0–60 cm). We used the regsubsets function in *leaps* (Lumley and Miller, 2009) in R with the exhaustive search method to determine the four best models per level of parameters. Leaps uses an efficient branch-and-bound algorithm to rapidly determine the best models. We then used Schwarz's Bayesian information criterion to identify the four best models. We report traditional parametric statistics for models. Residual analyses included visual confirmation that the assumptions of normality and homoscedasticity of residuals were not violated. We assessed multicollinearity among parameters in selected models with variance inflation factors. We also conducted an outlier analysis using Cook's distance (Cook and Weisberg, 1982) to see whether the model results changed, if the sample(s) with the highest influence on the particular model outcomes were excluded from the best MLR model. We calculated Cook's distance and excluded all samples with a Cook's distance greater than one while rerunning the particular model without them. This had no effect on significance levels.

3. Results

3.1. Responses to grazing treatments

When we compared conventional grazing to the average of the four other grazing treatments, we found that conventional management (summer and moderate grazing) tended to accumulate either similar or less SOC than all new grazing management treatments over the five-year study. For SOC stock of the full soil depth increment (0–60 cm), conventional management had 0.89 ± 0.31 kg × m^{-2} (±1 SE) less SOC than other grazing treatments (likelihood ratio test, $\chi^2 = 8.29$, df = 1, P = 0.004). For this and tests of other soil depth increments, nearly identical results were detected when SOC stocks were corrected for roots and rocks (data and results not shown) vs rocks (shown). In contrast, we found no appreciable difference (0.19 ± 0.13 kg of SOC per m^2) in SOC stocks of a surface soil depth increment (0–30 cm) ($\chi^2 = 2.07$, df = 1, P = 0.150). For SOC stocks of a standard mass of soil, there tended to be less (0.30 ± 0.16 kg) SOC in the top 0.4 t (dry soil) m^{-2} for conventional management when corrected for roots ($\chi^2 = 3.45$, df = 1, P = 0.063) but not when corrected for roots and rocks ($\chi^2 = 2.66$, df = 1, P = 0.103).

When we compared each new grazing practice to conventional management, we found no livestock grazing management treatment consistently changed SOC stocks relative to conventional management (Table 1). For SOC stocks of the full soil depth increment (0–60 cm), conventional management accrued 1.22, 1.02, and 0.85 kg of SOC per m^2 less than severe summer grazing, severe fall grazing, and no grazing, respectively (Table 1; Fig. 2B). We detected no other appreciable (P < 0.1) effects of treatments on other SOC stock metrics (Table 1).

3.2. Interrelationships between variables

We tested whether change in SOC stocks was explained by four hypothesized drivers of SOC accrual (i.e. soil mass, soil nitrogen, root biomass, plant species composition) (Fig. 1). For SOC stock by equivalent soil mass, change in SOC stocks was positively correlated with soil nitrogen (r = 0.56, Table 2). Moderate amounts of variation for change in SOC stocks (~30 %) was explained by multiple equally parsimonious multiple linear regression models (Table 3). Among the four best models, two soil predictor variables (nitrogen and soil mass) were commonly included (Table 3). Change in plant community composition (PC1, PC1 + 2) and root biomass were also included but in fewer models.

Table 1
Mean difference (±1 SE) in soil organic carbon (SOC) stock changes for conventional vs four alternative management types on four metrics of SOC stocks. For example, the second value ($-1.22 \pm 0.39^{0.002}$) indicates conventional management accumulated 1.22 kg × m^{-2} less SOC for the 0–60 cm soil depth increment than areas with severe summer grazing (P = 0.002, P-values as superscripts) over the duration of the experiment (2013–2018). Significant differences (α = 0.05) are in bold.

		Difference (kg × m^{-2}) in change in SOC (2013 to 2018) between conventional and alternative grazing management.			
Conventional	Alternative	$SOC_{0\text{-}30}$	$SOC_{0\text{-}60}$	$SOC_{0.4\ T}$	$SOC_{0.4\ T*}$
Summer, moderate	Summer, severe	$-0.21 \pm 0.17^{0.202}$	$\mathbf{-1.22 \pm 0.39^{0.002}}$	$-0.33 \pm 0.20^{0.107}$	$-0.28 \pm 0.19^{0.143}$
Summer, moderate	Fall, moderate	$-0.24 \pm 0.16^{0.141}$	$-0.49 \pm 0.39^{0.210}$	$-0.31 \pm 0.20^{0.126}$	$-0.30 \pm 0.19^{0.111}$
Summer, moderate	Fall, severe	$-0.04 \pm 0.17^{0.830}$	$\mathbf{-1.02 \pm 0.39^{0.010}}$	$-0.29 \pm 0.20^{0.160}$	$-0.19 \pm 0.19^{0.318}$
Summer, moderate	No grazing	$-0.27 \pm 0.17^{0.110}$	$\mathbf{-0.85 \pm 0.39^{0.031}}$	$-0.27 \pm 0.20^{0.183}$	$-0.21 \pm 0.19^{0.268}$

SOC = soil organic carbon stock (kg × m^{-2}) corrected for rocks for either 0–30 or 0–60 cm depth increments; $SOC_{0.4\ T}$ = soil organic carbon stock (kg SOC in the top 0.4 t [dry soil] m^{-2}) corrected for roots [standard metric by Gifford & Roderick (2003)]; $SOC_{0.4\ T*}$ = soil organic carbon stock (kg SOC in the top 0.4 t [dry soil] m^{-2}) corrected for roots and stones. Data are shown in Fig. 2.

Qualitatively similar results were found for a metric of SOC stocks for the full (0–60 cm) soil depth increment (Tables S1-2). Furthermore, root biomass was the top predictor and negatively associated with change in SOC stocks (r = −0.45, Table S1).

3.3. Importance of study design

By ignoring our 2013 data, we compared the results based on a Before-After Randomized Control Trial (BA-RCT) to a RCT where data are collected only after applying treatments. The BA-RCT design was able to detect several grazing effects (α = 0.05) on SOC stocks for the maximum soil depth increment (P ≤ 0.03; Table 1) that were not detected with a RCT design (P ≥ 0.22; Table S3). The RCT design also detected a treatment effect (P = 0.03; Table S3) on SOC stocks of an equivalent mass of soil that was not detected with the BA-RCT design (P = 0.16; Table 1).

4. Discussion

4.1. Responses to grazing treatments

Contrary to predictions, severe summer grazing had the largest positive effect on SOC stocks in the full soil depth increment and in our system. Other regional studies of longer duration (12–21 years) reported no difference in SOC stocks between light vs severe grazing treatments (0–60 cm) (Ingram et al., 2008; Schuman et al., 1999). Furthermore, a *meta*-analysis indicated severe grazing (i.e. beyond a system's carrying capacity) generally reduced SOC stocks (Abdalla et al., 2018). A possibility is that severe grazing has positive effects in the short-term (≤5 yr) and neutral or negative effects in the longer-term (≥10 yr) for temperate C_3 grasslands. Direct comparisons between our study and others, however, are complicated by key differences in methods (e.g. see section *4.3 Study design and statistical inaccuracy*). While severe grazing may increase SOC stocks in the short-term at our study site, it may also degrade grassland composition (Porensky et al., 2016), soil structure (Reinhart et al., 2019), and other ecosystem functions (Harrison et al., 2021; Reinhart and Vermeire, 2021).

The prevalence of positive effects (3 of 4 comparisons) of divergent grazing treatments (Table 1) on SOC stocks of the largest soil depth increment suggests some likely importance of management heterogeneity in our system. A positive effect of a new grazing practice on SOC

Table 2
Pearson product-moment correlations (r) for change in plant and soil properties over time in paddocks of a Northern mixed-grass prairie (n = 20) with four grazing treatments. For example, the first value (r = 0.56) indicates a positive correlation between log response ratios for SOC stocks for a standard mass of dry surface soil and soil nitrogen. In other words, an increase in SOC stocks from 2013 to 2018 was associated with increased soil nitrogen over the same time. In bold are correlation coefficients with significant p-values (*<0.05, **<0.01, ***<0.001).

	lrr_SOC	lrr_N30	lrr_N60	lrr_root30	lrr_root60	lrr_soil30	lrr_soil60
lrr_N30	**0.56***						
lrr_N60	0.38	**0.70*****					
lrr_root30	−0.32	−0.31	−0.18				
lrr_root60	−0.39	−0.32	−0.17	**0.98*****			
lrr_soil30	0.01	**−0.47***	**−0.65****	−0.17	−0.22		
lrr_soil60	0.22	−0.03	**−0.55***	−0.21	−0.26	**0.64****	
PC1	−0.11	0.35	0.14	−0.15	−0.16	−0.28	−0.05
PC1 + 2	−0.41	−0.09	0.09	0.18	0.18	−0.23	−0.43

Log response ratios describe change in properties pre- (2013) and post-treatment (2018). Abbreviations: lrr_SOC (log response ratio for kg SOC in the top 0.4 t (dry soil) m^{-2} corrected for roots and stones), lrr_N30 (log response ratio of soil nitrogen concentration for 0 to 30 cm), lrr_N60 (log response ratio of soil nitrogen concentration for 0 to 60 cm), lrr_root30 (log response ratio of root biomass kg × m^{-2} for 0 to 30 cm), lrr_root60 (log response ratio of root biomass kg × m^{-2} for 0 to 60 cm), lrr_soil30 (log response ratio of total dry weight [g] of mineral soil [free of rocks and roots] per core for 0 to 30 cm), lrr_soil60 (log response ratio of total dry weight [g] of mineral soil [free of rocks and roots] per core for 0 to 60 cm), PC1 (change in principle component 1 of plant species composition of a paddock between 2014 and 2018), and PC1 + 2 (change in principle component 2 between 2014 and 2018 summed with PC1).

Table 3
Four best multiple regression models, based on Schwarz Bayesian Criterion scores (BIC), to explain change in SOC stocks (kg SOC in the top 0.4 t (dry soil) m^{-2} corrected for roots and stones) between 2013 and 2018 of grazed paddocks (n = 20).

Models	Independent variables	BIC	*t*-value	F	P	Adjusted R^2
1	lrr_soil60	–	2.201	–	0.044	–
	lrr_root60	–	−0.785	–	0.4447	–
	lrr_N60	–	2.806	–	0.0133	
	PC1	–	−1.076	–	0.2988	
	total	−2.668	–	3.382	0.037	0.334
2	lrr_soil60	–	2.817	–	0.0119	–
	lrr_N60	–	3.284	–	0.0044	–
	total	−2.645	–	6.132	0.0099	0.351
3	lrr_soil60	–	1.534	–	0.146	–
	lrr_root60	–	−0.645	–	0.529	
	lrr_N60	–	2.512	–	0.024	
	PC1 + 2	–	−1.259	–	0.227	
	total	−1.853	–	3.57	0.031	0.351
4	lrr_root60	–	−1.597	–	0.129	–
	lrr_N60	–	1.538	–	0.142	–
	total	−1.876	–	2.972	0.0782	0.172

Log response ratios describe change in properties pre- (2013) and post-treatment (2018). Abbreviations: lrr_N60 (log response ratio of soil nitrogen concentration for 0 to 60 cm), lrr_root60 (log response ratio of root biomass kg × m^{-2} for 0 to 60 cm), lrr_soil60 (log response ratio of total dry weight [g] of mineral soil [free of rocks and roots] per core for 0 to 60 cm), PC1 (change in principle component 1 of plant species composition of a paddock between 2014 and 2018), and PC1 + 2 (change in principle component 2 between 2014 and 2018 summed with PC1). Significance of linear model was tested with ANOVA. Numerator and denominator degrees of freedom, respectively, were 2,17 for model 1; 3,16 for models 2 and 4; and 1,18 for model 3.

stocks is expected to gradually plateau as a new stable equilibrium is reached. Management heterogeneity (e.g. changes to grazing management over time) may allow SOC stocks to attain new maximum levels (e.g. additive effects) relative to areas where only a single grazing treatment is annually applied. In contrast to the continuous application of severe grazing, management heterogeneity may help to increase SOC stocks while maintaining ecological sustainability (e.g. forage production, plant community composition, hydrologic function). Unfortunately, nearly half of the primary literature consists of experiments with overly simplistic treatments (e.g. effects of traditional grazing vs no grazing) (review by Reinhart et al., 2022b) and tells us little about the importance of management heterogeneity. More research is needed to understand whether unique sequences (or durations) of grazing treatments may accrue different levels of SOC and have fewer unintended consequences.

A plan to fight climate change is to adopt land-use practices that increase SOC stocks by 0.4 % per year (e.g. Chambers et al., 2016; Minasny et al., 2017; Soussana et al., 2019). This study is one of the few that explicitly measured change in SOC stocks over time, which is an emerging priority (FAO, 2019; Reinhart et al., 2022b). Furthermore, severe summer grazing increased SOC stocks (0–60 cm) by 1.8 % per year over a 5 yr period while conventional management (moderate summer grazing) decreased SOC stocks by 2.5 % per year in our system. We are unsure why stocks decreased for the conventional management treatment. This decrease may indicate natural variation around a stable state. A further consideration is that areas with a relatively high level of SOC stocks at the start of the experiment likely had a smaller capacity for SOC accrual than areas with a lower initial SOC stock (Arndt et al., 2022; Stewart et al., 2007). Another possibility is that our conventional management paddocks lacked pasture-level realism. For example, the paddocks were grazed by few livestock for 3–5 days a year whereas the pasture where these paddocks occur was grazed by tens of livestock for two or more months a year prior to study. Schönbach et al. (2012) conducted the only other related study that measured change to SOC stocks, and they measured only the topsoil (0–5 cm) and reported no appreciable grazing treatment effects.

Ours is the first study to test effects of grazing season on SOC stocks. Severe fall grazing increased SOC stocks in the 0–60-cm zone by 0.8 % per year. Despite the lack of studies with grazing season treatments, grazing dormant forages in the fall (and winter) is a common practice (McGeough et al., 2017; Wyffels et al., 2019).

Similar to severe fall grazing, grazing exclusion also increased SOC stocks of full depth increment (0–60 cm) by 0.5 % per year. This was unexpected because a study of surface soils (0–20 cm depth) at the same research station reported greater SOC stocks in grazed than not grazed areas (Reinhart et al., 2021). This discrepancy likely relates to differences in sampling depths, numbers of years without grazing, site-to-site variation, and whether pre-treatment data were collected. Furthermore, soil density decreases as the number of years without grazing increases (Reinhart et al., 2021), thereby indicating a gradual reversal of past biocompaction by livestock. This change in soil density coupled with the positive relationship between bulk density and SOC stocks (e.g. Table S2) may explain why no grazing increases SOC stocks in the short-term but decreases stocks in the longer term (Reinhart et al., 2021). Other studies in our region reported no difference in SOC metrics of areas with conventional management vs no grazing (Biondini et al., 1998; Derner et al., 2019; Henderson et al., 2004; Ingram et al., 2008; Shrestha and Stahl, 2008).

While we found treatment effects on SOC stocks of the full sampling depth interval, we found no appreciable effects of treatments on SOC stocks (SOC mass × area^{-1}) for the shallow depth increment (0–30 cm, Fig. 1A) and those based on a standard mass of soil (kg SOC in the top 0.4 t [dry soil] m^{-2}; Fig. 1C,D) which was designed to describe a similar soil depth increment (Gifford and Roderick, 2003). We interpret that the failure to detect grazing treatment effects of surface soils relative to the maximum soil depth increment is further evidence for increasing the standard depth (30 cm) (Intergovernmental Panel on Climate Change, 2006) for sampling soil of grazing lands (FAO, 2019).

4.2. Indicators of SOC stock change

Many are interested in easily measured proxy variables that are correlated with SOC stocks as well as variables that might represent causal mechanisms explaining accrual of SOC (Jackson et al., 2017; Wiesmeier et al., 2019). Potential indicators of SOC stocks include lignin-modifying enzymes (Roy and Bagchi, 2022), soil nitrogen (Piñeiro et al., 2009), root biomass (Wilson et al., 2018), soil density (Reinhart et al., 2021), primary productivity (Henderson et al., 2015), and plant species composition (Bagchi and Ritchie, 2010). We tested for relationships between change in SOC stocks and several frequently reported variables (Fig. 1). Most variables explained some variation in SOC stocks. Soil nitrogen (N) was a top predictor and positively associated with change in SOC stocks. Others noted an increase in soil N was associated with decreasing activity of lignin-modifying enzymes and increasing SOC (Chen et al., 2018; Roy and Bagchi, 2022). Exactly how grazing affects soil N is unclear but may relate to excreta (dung and urine), shifts in plant material inputs, shifts in soil physical properties, shifts in N content of soil microbial biomass, and shifts in nitrogen fixing bacteria (Mariotte et al., 2017; Ritchie and Raina, 2016; Sankaran and Augustine, 2004; Schrama et al., 2013).

We also found a negative relationship between change in root biomass and change in SOC stocks (e.g. Table S1). In contrast, Wilson et al. (2018) identified a positive relationship between root biomass and SOC stocks for a subtropical grassland. Various factors (e.g. soil moisture, nitrogen) likely regulate whether root inputs increase or decrease (e.g. prime microbial mineralization) SOC stocks (Dijkstra et al., 2021; Huo et al., 2022). Because of these opposing dynamics, roots are an unreliable indicator of SOC stocks.

Soil structural properties (e.g. soil bulk density, aggregate stability) may also affect SOC stocks (Oades, 1984; Six et al., 2002; Six and Paustian, 2014). While SOC stocks for an equivalent mass of surface soil controlled for the mass of surface soil (ca. 30 cm depth), soil mass across the full soil depth increment (60 cm depth) was positively associated with change in SOC stocks for a standard mass of soil. Others reported a positive effect of soil bulk density on SOC stocks (Deurer et al., 2012; Elschot et al., 2015) suggesting that an increase in bulk density may reduce SOC mineralization and promote SOC accrual. Furthermore after 2 years of applying treatments, Reinhart et al. (2019) reported a negative effect of severe summer grazing (and all fall grazing treatments) relative to no grazing on the stability of small-sized macroaggregates (0.25–1 mm) of topsoil (0–10 cm depth). The differential effects of grazing treatments on SOC stocks vs soil aggregate stability implies a negative association between the two soil properties in a grazing land context. This is paradoxical because soil aggregates are thought to protect soil organic matter and SOC from mineralization (e.g. Oades, 1984; Six and Paustian, 2014). We incorrectly predicted that SOC stocks of fine soil should be greater where soils are less compacted (e.g. lower bulk density) and contain a greater percentage of water-stable aggregates. Similar to root inputs, soil structural properties (i.e. bulk density, aggregate stability) in grazed systems may have divergent associations with SOC stocks. Our findings are a further indication of the uncertainty over the drivers of grazer effects on SOC stocks (e.g., Roy and Bagchi, 2022).

4.3. Study design and statistical inaccuracy

Since most C ranching experiments are little-replicated without pre-treatment data (Reinhart et al., 2022b), the reliability of their results may depend on pre-existing differences (Arndt et al., 2022). For example, analyses without pre-treatment data (and little to no treatment replication) cannot differentiate intrinsic differences among experimental units (i.e. paddocks) from differences caused by grazing treatments. To gauge the dependence of results on pre-treatment data, we compared results for our Before-After Randomized Control Trial (BA-RCT; gold standard) to a RCT (silver standard) (Reinhart et al., 2022b). By controlling for pre-existing differences, the BA-RCT design detected several grazing effects on SOC stocks ($\alpha = 0.05$) for the largest soil depth increment (Table 1) not detected by our RCT (Table S3) or other longer duration RCT studies (Ingram et al., 2008; Schuman et al., 1999). The RCT design also detected a treatment effect on SOC stocks of a standard mass of soil (Table S3) that was not detected with the BA-RCT design (Table 1). When experiments have little treatment replication, unreliable results are more likely for a RCT design than a BA-RCT.

5. Conclusion

Unfortunately, the most promising grazing treatment (i.e. severe grazing) for short-term (5 yr) accrual of SOC in our system will likely have the most detrimental effects on other ecosystem functions. Unclear is the grazing practice that will reliably and sustainably maximize the rate of SOC accrual for a specific climate and timeline. Management heterogeneity may be key to sustainably accruing maximum SOC. Our findings counter the prevailing view that plant material inputs are the primary driver of SOC accrual in grazing lands. Short-term SOC accrual was mainly associated with indicators of decreased SOC mineralization (C outputs) and was negatively (not positively) associated with plant material inputs. Unfortunately, many assume a positive functional relationship exists and estimate “soil C sequestration” from plant material (e.g. NDVI) and other types of data—groups include C ranching modelers (Henderson et al., 2015), C ranching offset companies (Deriemaeker et al., 2018), and advocates of regenerative ranching (e.g. Gosnell et al., 2020). Our novel C ranching dataset also enabled comparisons of results for analyses with vs without pre-treatment data. Our findings suggest that the most common types of study designs (review by Reinhart et al., 2022b) may yield unreliable results. For C ranching and C offset projects to succeed at offsetting carbon dioxide emissions, a next generation of studies (i.e. experimental designs, treatments) over broader spatio-temporal scales are urgently needed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The following data set has been archived (Reinhart et al., 2022a): (1) CSV data files; (2) R script of data wrangling and analyses; (3) R session info; and (4) readme with descriptions of files, workflow, and metadata.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geoderma.2022.116061.

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